

$$\begin{bmatrix} 1 & 0 & 0 & -2 & -3 \\ 0 & 1 & 1 & 0 & 0 \\ 0 & 0 & 1 & 3 & 1 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

$$\text{Row 3} = \text{Row 3} - \text{Row 2}$$

$$\begin{bmatrix} 1 & 0 & 0 & -2 & -3 \\ 0 & 1 & 1 & 0 & 0 \\ 0 & -1 & 0 & 3 & 1 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

The corresponding equations are:

$$\textcircled{1} x_1 - 2x_4 = -3 \quad \textcircled{2} x_2 + x_3 = 0$$

$$\textcircled{3} -x_2 + 3x_4 = 1 \quad \textcircled{4} 0 = 0$$

If $x_2 = -1$, from $\textcircled{3}$, $x_4 = 0$. From $\textcircled{2}$, $x_3 = 1$. From $\textcircled{1}$, $x_1 = -3$.

$\therefore$ The equations are consistent since at least one solution exists. $\square$

17) Do the three lines $x_1 - 4x_2 = 1$, $2x_1 - x_2 = -3$, and $-x_1 - 3x_2 = 4$ have a common point of intersection? Explain.

Soln: The given three lines have a common point of intersection if and only if there is a unique value of x_1 and x_2 that satisfies the eqns.

Augmented matrix: $\begin{bmatrix} 1 & -4 & 1 \\ 2 & -1 & -3 \\ -1 & -3 & 4 \end{bmatrix}$ $\text{Row 3} = \text{Row 3} + \text{Row 1}$

$$\begin{bmatrix} 1 & -4 & 1 \\ 2 & -1 & -3 \\ 0 & -7 & 5 \end{bmatrix}$$

$$\begin{aligned} \text{Row 2} &= \text{Row 2} - 2 \times \text{Row 1} \\ \text{Row 2} &= \text{Row 2} \times \frac{1}{7} \\ \text{Row 3} &= \text{Row 3} \times \frac{1}{7} \end{aligned}$$

$$\begin{bmatrix} 1 & -4 & 1 \\ 0 & 7 & -5 \\ 0 & -1 & 5 \end{bmatrix}$$

$$\begin{bmatrix} 1 & -4 & 1 \\ 0 & 1 & -5/7 \\ 0 & 1 & -5/7 \end{bmatrix}$$

$$\text{Row 1} = \text{Row 1} + 4 \times \text{Row 2}$$

$$\begin{bmatrix} 1 & 0 & -13/7 \\ 0 & 1 & -5/7 \\ 0 & 1 & -5/7 \end{bmatrix}$$

$$\begin{aligned} x_1 &= -13/7 \\ x_2 &= -5/7 \\ x_3 &= -5/7 \end{aligned}$$

$\therefore$ There is a common point of intersection. $\square$

18) Do the three planes $x_1 + 2x_2 + x_3 = 4$, $x_2 - x_3 = 1$, and $x_1 + 3x_2 = 0$ have at least one common point of intersection? Explain.